

STA 35C: Statistical Data Science III

Lecture 17: Regularization Methods (cont'd) & Multiple Testing

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Announcement

Midterm 2 on Fri, May 15 (1:10 pm–2:00 pm in class)

- **Arrive early:** The exam starts at 1:10 pm and ends at 2:00 pm sharp
- **One hand-written cheat sheet:** Letter-size (8.5"×11"), double-sided, brief formulas/notes
- **Calculator:** A simple (non-graphing) scientific calculator is allowed
- **No other materials** beyond the single cheat sheet (no textbooks, etc.)
- **SDC accommodations:** Confirm scheduling with AES online ASAP

Advice for preparation:

- Primary coverage: Lectures 12–19 (including next Wed)
- Core concepts from earlier may be assumed, especially regression and model assessment
- Two practice midterms and brief solution keys are posted on course webpage
- Office hours next week:
 - Instructor: Wed, 3:30–4:30 pm
 - TA: Tue 12:30–2:30pm

Agenda

- **Regularization: ridge vs. lasso**
 - Recap: Regularization, ℓ_2 penalty (ridge regression), and selecting λ
 - Lasso: ℓ_1 -penalized regression and variable selection
 - Geometric intuition and comparison between ridge vs. lasso
- **Multiple hypothesis testing:** Motivation
 - Why ordinary single-hypothesis testing may fail in large-scale settings
 - Preview: Type-I error inflation and how to control it

Recap: Why regularization?

Challenges: Least squares estimates can

- be unstable when predictors are highly correlated or the data are noisy;
- be non-unique when $p > n$ or the design matrix is rank-deficient;
- use all predictors even when the true relationship is sparse.

Regularization can stabilize estimation by adding a penalty term: with $\lambda \geq 0$,

$$\hat{\beta}_{\lambda} \in \arg \min_{(\beta_0, \beta_1, \dots, \beta_p)} \left\{ \underbrace{\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2}_{\text{RSS}} + \lambda \underbrace{R(\beta_1, \dots, \beta_p)}_{\text{penalty}} \right\}$$

- The penalty shrinks coefficients, trading some bias for lower variance

Two popular choices:

- **Ridge:** $R(\beta_1, \dots, \beta_p) = \sum_{j=1}^p \beta_j^2$
- **Lasso:** $R(\beta_1, \dots, \beta_p) = \sum_{j=1}^p |\beta_j|$

Recap of ridge: Regularization reduces variance via shrinkage

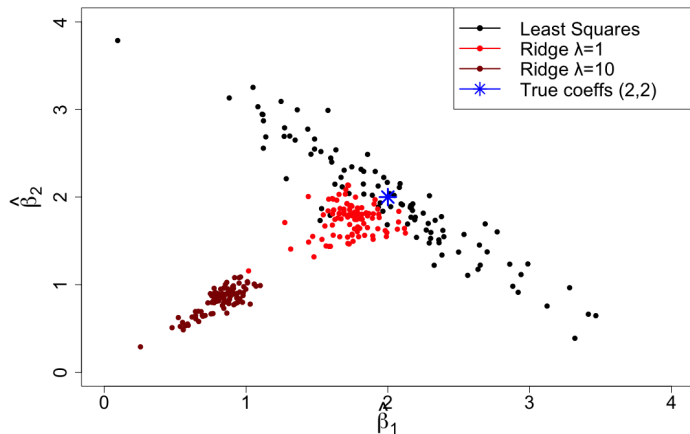


Figure: Scatter plots of 100 least squares estimates (**black**) vs. ridge estimates for $\lambda = 1$ in **red** and $\lambda = 10$ in **dark red**. As λ grows, the estimates cluster more tightly (lower variance) but shift away from the true value (**blue star**), indicating increased bias.

Recap of ridge: Illustration of shrinkage with 1D example

In the simplified setting with $n = p = 1$ without intercept, ridge solves for $\lambda \geq 0$:

$$\hat{\beta}_{\lambda}^R \in \arg \min \left\{ (y - x\beta)^2 + \lambda\beta^2 \right\}$$

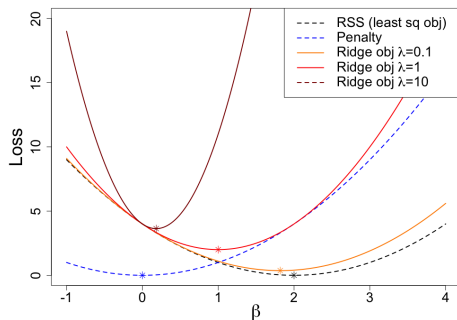


Figure: As λ grows, $\hat{\beta}_{\lambda}^R$ shrinks toward 0 for fixed (y, x) ($y = 2$, $x = 1$).

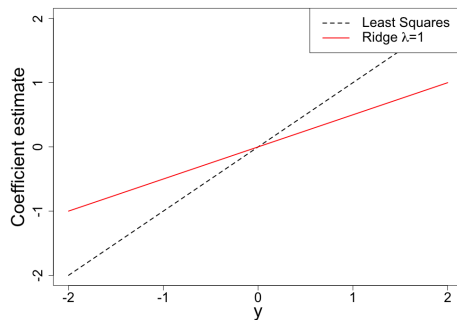


Figure: For each y , $\hat{\beta}_{\lambda}^R$ is smaller than the LS estimate y/x in magnitude, when $\lambda > 0$.

Recap of ridge: Visualization of bias-variance tradeoff

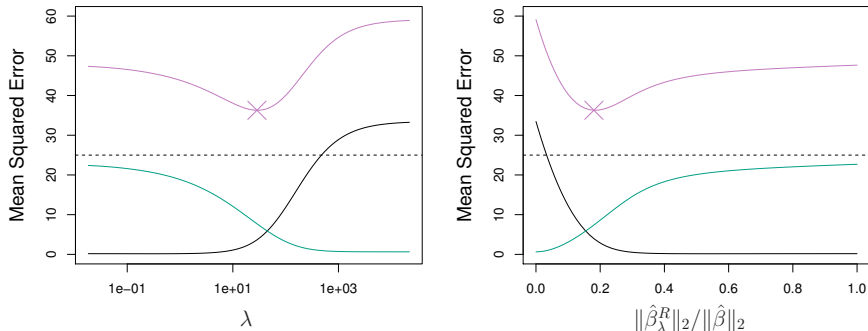


Figure: Bias-variance tradeoff in ridge for a simulated data set ($p = 45$, $n = 50$). Shown are squared bias (**black**), variance (**green**), and test MSE (**purple**) vs. λ [JWHT21, Figure 6.5].

- At $\lambda = 0$, there is no bias but high variance
- Increasing λ *significantly* reduces variance at the cost of *slightly* higher bias
- Eventually, added bias overtakes the benefit of reduced variance

The lasso: Formulation

The lasso: Find $\hat{\beta}_0, \dots, \hat{\beta}_p$ that minimize

$$\underbrace{\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2}_{\text{RSS}} + \underbrace{\lambda \sum_{j=1}^p |\beta_j|}_{\text{penalty}}$$

- $\lambda \geq 0$ is a tuning parameter
- Each choice of λ gives a different set of Lasso estimates $\hat{\beta}_\lambda^L$

Remarks:

- No penalty on β_0 (the intercept)
- As $\lambda \rightarrow 0$, lasso approaches the least-squares solution when it is uniquely defined
- As λ grows, many β_j shrink toward zero, and some exactly become 0

Unlike ridge, the lasso can yield exact zero estimates $\implies$ **variable selection!**

Lasso: Illustration with 1D example

In the setting with $n = p = 1$ without intercept, the lasso solves, for $\lambda \geq 0$,

$$\hat{\beta}_{\lambda}^L \in \arg \min_{\beta} \left\{ (y - x\beta)^2 + \lambda|\beta| \right\}$$

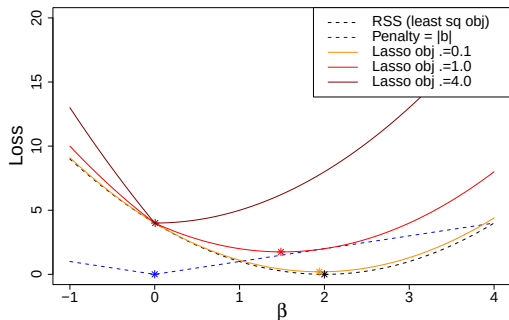


Figure: As λ grows, $\hat{\beta}_{\lambda}^L$ shrinks more aggressively; small $|y|$ can yield $\beta = 0$ ($y = 2$, $x = 1$ fixed).

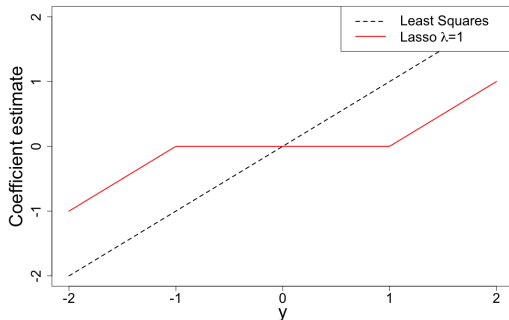


Figure: For sufficiently small $|y|$, β is exactly 0. This “thresholding” underlies variable selection.

Lasso: Regularization reduces variance, but...

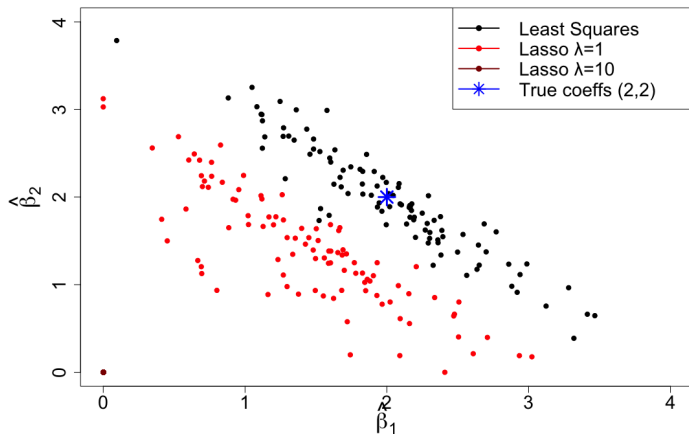


Figure: Scatter plots of 100 least squares estimates (**black**) vs. lasso estimates for $\lambda = 1$ in **red** and $\lambda = 10$ in **dark red**. Lasso can aggressively shrink or zero-out coefficients, but the variance reduction is less uniform than ridge. The shift from the true (**blue star**) may or may not be worth it.

Lasso: Regularization enables variable selection

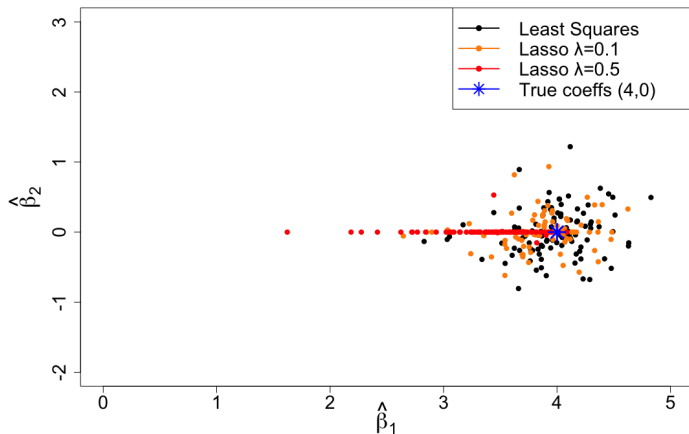


Figure: Scatter plots of 100 least squares estimates (black) vs. lasso estimates for $\lambda = 0.1$ in orange and $\lambda = 0.5$ in red. If the true $\beta_2 = 0$ (blue star), lasso can correctly select the significant variable (X_1), while suppressing noise and driving estimates to zero for X_2 , thereby capturing the “sparse” true associations.

The lasso: Credit dataset example

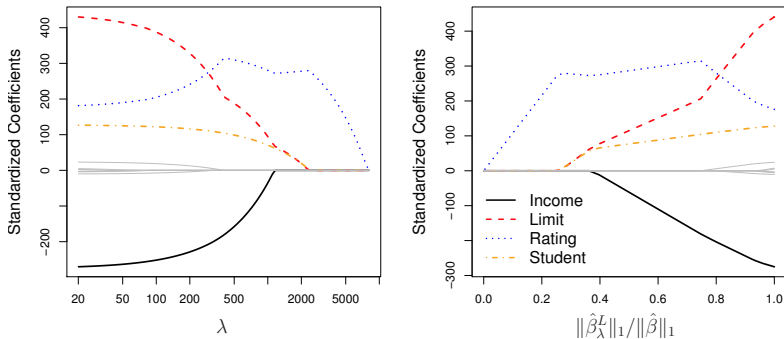


Figure: Standardized lasso coefficients for Credit, plotted vs. λ and the shrinkage level $\|\hat{\beta}_\lambda^L\|_1 / \|\hat{\beta}\|_1$ [JWHT21, Figure 6.6].

- Lasso can force some coefficients to zero as λ increases
- Achieves *variable selection* directly (predictors with $\hat{\beta}_j^L = 0$ are excluded)

The lasso: Selecting λ

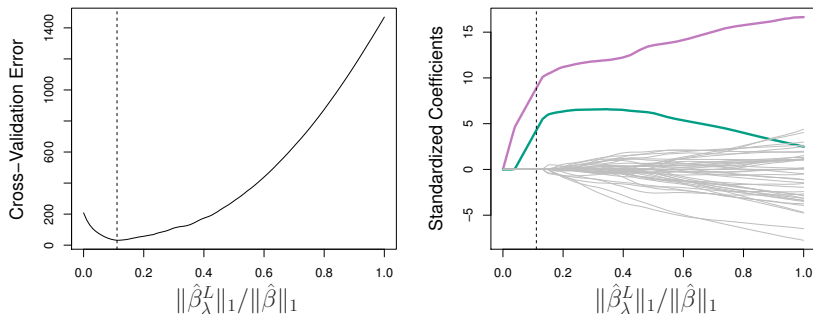


Figure: **Left:** Ten-fold cross-validation MSE for the lasso, applied to the sparse simulated data. **Right:** The corresponding lasso coefficient estimates, with the two *signal* variables in color and the *noise* variables in gray. The vertical dashed line indicates the fit that minimizes the cross-validation error [JWHT21, Figure 6.13].

- The lasso cleanly separates two *signal* variables from *noise* variables
- In contrast, standard least squares (far right, with $\|\hat{\beta}_\lambda^L\|_1 / \|\hat{\beta}\|_1 = 1$) only identifies the **purple** variable without discarding the noise predictors

The lasso: Example with a toy dataset (R script)

- **Setup:**
 - $n = 5, p = 2$
 - Compare lasso vs. OLS at $\lambda = 1$
- **Objective:** See how lasso can shrink coefficients to zero

```
# Toy data: 5 obs, 2 predictors
df <- data.frame(
  x1 = c(1,2,3,4,5),
  x2 = c(2,1,3,1,2),
  y = c(2,2.5,6,4,6.5)
)
```

```
# OLS fit
ols_fit <- lm(y ~ x1 + x2, data=df)
cat("OLS Coeffs:\n", coef(ols_fit), "\n")

# Run install.packages("glmnet") once if needed
library(glmnet)

# Prepare X, y
X <- as.matrix(df[, c("x1","x2")])
y <- df$y

# Lasso with alpha=1, lambda=1
lasso_fit <- glmnet(X, y,
                    alpha=1,
                    lambda=1,
                    intercept=TRUE,
                    standardize=TRUE)
cat("Lasso Coeffs (lambda=1):\n",
    as.matrix(coef(lasso_fit)), "\n")
```

Pop-up quiz: Lasso

Suppose a lasso model is fit for several values of λ .

Question: What generally happens as λ increases?

- A) The penalty weakens, so coefficients move closer to least squares.
- B) Coefficients are increasingly shrunk, and some may become exactly zero.
- C) Training RSS must decrease because the model becomes more flexible.
- D) The intercept is penalized more strongly than the other coefficients.

Answer: B. Increasing λ strengthens the ℓ_1 penalty; lasso shrinks coefficients and can set some exactly to zero, enabling variable selection.

(Optional¹) Alternative formulation: Constrained form

Ridge and lasso can be expressed as equivalent *constrained* optimization problems:

$$\begin{aligned} \text{(Ridge)} \quad & \text{minimize}_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s_{\lambda} \\ \text{(Lasso)} \quad & \text{minimize}_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s'_{\lambda} \end{aligned}$$

- For each $\lambda \geq 0$, there exist corresponding $s_{\lambda}, s'_{\lambda}$ such that solving the above problems yield the same ridge/lasso regression coefficient estimates
- Geometrically: feasible region is an ℓ_2 -ball for ridge or ℓ_1 -ball for lasso

¹That is, it is good to know, but its mathematical details will not be asked in the exams

The lasso prefers “spiky” solutions

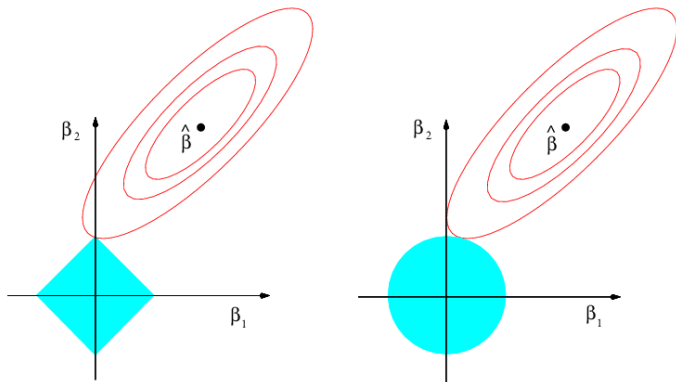


Figure: Contours of the RSS (red ellipses) and the feasible sets (cyan areas). **Left:** For lasso, the constraint $\|\beta\|_1 \leq s$ (a diamond shape) can yield corner solutions having exact zeros. **Left:** For ridge, the constraint $\|\beta\|_2^2 \leq s'$ is round, so typically yielding no exact zeros [JWHT21, Figure 6.7].

Comparison of ridge vs. lasso

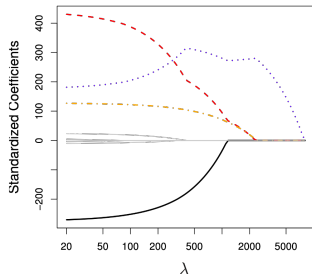
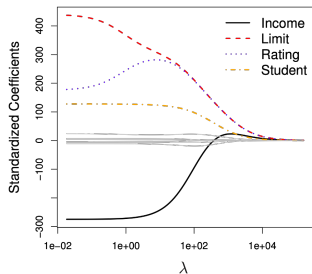


Figure: Standardized ridge (left) and lasso (right) coefficients on **Credit** dataset, plotted vs. λ [JWHT21, excerpted from Figures 6.4 & 6.6].

Ridge:

- More stable under collinearity
- Typically no exact zeros
- Often simpler closed-form solution

Lasso:

- Possibly less stable under correlated predictors
- Produces zero coefficients (variable selection)
- More interpretable if many X_j are irrelevant

Regularization: Summary

- **Why regularization?**

- Remedy high variance or ill-posedness, especially when $p \approx n$ or $p > n$
- Potentially yield simpler, more interpretable models (esp. lasso)

- **How?** Add a penalty

- Ridge: $\sum_{j=1}^p \beta_j^2$ shrinks all β_j stably, rarely yielding exact zeros
- Lasso: $\sum_{j=1}^p |\beta_j|$ can drive some β_j to 0, enabling variable selection
- Tuning parameter λ typically selected via cross-validation

- **Ridge vs. Lasso:**

- Ridge is stable under collinearity and has simpler closed-form solutions
- Lasso can yield sparse solutions (some $\beta_j = 0$)
- Neither strictly dominates: test performance depends on the data
→ usually do cross-validation to choose

Pop-up quiz: Regularization

Question: Which statement is *false* regarding ridge and lasso?

- A) Ridge often works well when many predictors have modest effects or are highly correlated.
- B) Lasso can set some coefficients exactly to zero, giving built-in variable selection.
- C) Once λ is chosen by cross-validation, ridge will always outperform lasso in test MSE.
- D) Both ridge and lasso use a tuning parameter λ that controls the amount of shrinkage.

Answer: C. Neither method uniformly dominates; after tuning, the better test performance depends on the data-generating structure.

Multiple hypothesis testing: Motivation

Recall single-hypothesis testing:

- For each predictor X_j , test $H_0 : \beta_j = 0$
- Reject H_0 if $p < \alpha$ (e.g., $\alpha = 0.05$); Type I error rate = α for *one* test
 - **Type I** (False positive): Null is true, but we reject
 - **Type II** (False negative): Null is false, but we fail to reject

Modern data analysis often tests **many variables** (or features) simultaneously

- We want to identify which predictors are “significant” among many candidates

Examples:

- Testing thousands of genes/biomarkers for disease association
- Testing many (possibly high-dimensional $p > n$) predictors for stock price forecasting

Problem: Merely applying ordinary tests to each predictor can yield many false positives

Multiple hypothesis testing: Illustration

Coin-flip analogy:

- Testing *fairness* of a coin: $H_0 : p = 0.5$
- Suppose 1,024 fair coins are each flipped 10 times
- For any one coin,

$$P(10 \text{ heads}) = \left(\frac{1}{2}\right)^{10} = \frac{1}{1024}.$$

- Standard two-sided test on that single coin gives p -value below 0.002
- However, seeing one coin with 10 heads (across 1024 tried) is NOT surprising!

Key points:

- With many tests, extreme results can happen just by chance
- We must account for that when claiming “significance”

Multiple hypothesis testing: Challenges

Setting:

- Suppose we have m predictors to test simultaneously
- Each test has a per-hypothesis Type I error rate $\alpha > 0$

Problem:

- With m tests, we have m chances for false positives
- If all m nulls are true and tests are independent,

$$P(\text{at least one false rejection}) = 1 - (1 - \alpha)^m,$$

which can be large as m grows

- e.g. at $m = 20$ and $\alpha = 0.05$, we expect ≈ 1 false positive on average

How to address?

- Requiring $p < 0.05$ for each *does not* guarantee a $\leq 5\%$ chance of *any* false positive
- We need **multiple-comparison corrections** (next Lecture)
 - *Family-Wise Error Rate (FWER)* ensures probability of *any* false positive is $\leq \alpha$
 - *False Discovery Rate (FDR)* limits the *proportion* of false positives among all rejections

Pop-up quiz: Multiple testing

Suppose we test $m = 20$ independent null hypotheses at level $\alpha = 0.05$, and suppose all null hypotheses are actually true.

Question: Which statement is most accurate?

- A) The probability of at least one false rejection is exactly 0.05.
- B) The expected number of false rejections is $20 \cdot 0.05 = 1$.
- C) The probability of at least one false rejection decreases as m increases.
- D) Multiple testing is not a concern if each individual test uses $\alpha = 0.05$.

Answer: B. If all nulls are true, each test has false-rejection probability 0.05, so the expected number of false positives is $m\alpha = 20 \times 0.05 = 1$.

Wrap-up & next steps

Regularization:

- Ridge (ℓ_2 penalty) is stable under correlated predictors
- Lasso (ℓ_1 penalty) can set some coefficients exactly to zero (variable selection)
- Typically pick λ via cross-validation

	Ridge	Lasso
Penalty	$\sum_{j=1}^p \beta_j^2$	$\sum_{j=1}^p \beta_j $
Effect	shrinks coefficients	shrinks and can set to zero
Variable selection	No	Yes
Often useful when	many small/moderate effects	only some predictors matter
Tuning	λ typically chosen by cross-validation	

Multiple hypothesis testing:

- Single-hypothesis framework can fail when m is large
 - Probability of at least one Type I error can be quite large
- We need corrections for controlling false positives

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.